Variable Speed Limits using Safe Soft Actor-Critic Reinforcement Learning for Traffic Flow Optimization of Long Highway Networks

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Abstract

Variable Speed Limits (VSL) is a technique of optimizing traffic flow, by dynamically changing speed limits, based on current traffic conditions. After going over past solutions, their strengths and limitations, this proposal aims to combine the best approaches among them, and ultimately contribute to reducing congestion, and improving safety. Reinforcement learning is used – specifically the soft actor-critic algorithm, simulated on METANET traffic model, and applied to real traffic data from the city of Dubai. A safety metric is also added.

1. INTRODUCTION

Variable Speed Limits (VSL) is a type of Intelligent Traffic Solution (ITS) that incorporates current traffic conditions into consideration to output appropriate limits to support traffic flow. These limits are meant to be displayed on overhead Variable Message Signs (VMS) to inform drivers. This proposal details the importance of this project, previous methods

1.1 Problem Definition

Traffic is a problem every urban city faces today. A recent report stated that employees spend hours equivalent to 60 working days a year commuting the Dubai-Sharjah route [4], and in some regions it exceeds 150. Conservative estimates suggest that increasing the average speed of private car journeys by 1 km/h and public transport by 0.5 km/h could reduce journey times and operating costs, amounting to savings equivalent to 0.1% of the GDP. Idling in traffic jams negatively affects quality of life - increasing traffic fatigue, reducing rest time, and lowering labour productivity. The more congestion that drivers experience, the more frustrated they become, leading to cases of road rage. Additionally, congestion also leads drivers to stop paying attention to the road. Once a driver gets into the pattern of moving a couple of feet every few minutes, they stop paying attention to the road as much. It not only increases travel times, but also contributes to fuel consumption in billions of litres, and greenhouse gas emissions in megatons per city. Multiple studies have shown that high speeds, traffic breakdowns (slowing down and accelerating again) and congestion emit higher levels of these gases, especially nitrogen dioxide. Higher traffic densities contribute to higher temperature in urban areas due to vehicle emissions and asphalt heating. These individual emissions can build up to 15% of the city’s total emissions [2], [11]. Traffic congestion is closely connected to Sustainable Development Goal (SDG) 11, which aims to promote sustainable cities and communities. Traffic congestion also significantly affects SDG 9, which focuses on Industry, Innovation, and Infrastructure. Efficient and resilient transportation infrastructure & solutions like this project are the key to this. Congestion has direct health implications, impacting SDG 3, which focuses on Good Health and Well-being. The heightened pollution caused by traffic congestion deteriorates air quality, leading to respiratory issues and other health problems for urban residents [12].

1.2 The Solution

The ideal scenario is perfect coordination – uniform speed, with equal distance between vehicles. It is difficult to achieve this in the real world, where just one vehicle’s lane change or slowing would make the many behind do so as well, each of them having different reaction times, creating ‘phantom’ intersections, i.e. slowing down/stopping & resuming behavior (also known as shockwaves) seen at intersections. Traffic jams don’t always require bottlenecks to form; they can emerge purely from the interactions between vehicles, which destabilize the flow. Contributing factors include tailgating (driving too closely to the car ahead), erratic reactions to speed changes, slow overtaking by larger vehicles, and queue-jumping. Additionally, vehicles merging from on-ramps increase the strain on traffic flow, particularly during demand surges. Disruptions like accidents or blockages exacerbate the situation by creating congestion that stretches far beyond the original incident. Traffic instability can stem from both predictable causes, such as driver behaviors, and random variations, underlining the complexity of maintaining steady and efficient traffic flow [1], [6].

1.3 Approaches to solutions

Many studies have used similar models & techniques for ramp metering [9], but VSL was chosen over this for the following reasons:

- VSL can have a potentially network-wide impact, while ramp metering is implemented at a point.
- Ramp metering is reactive, and VSL can be made to take actions in advance.
- Ramp metering does not directly affect speed variations & prevention of traffic breakdowns, involving safety & environmental aspects.

Traffic Light Controllers (TLC) are also widely studied for the same problems, and have achieved similar goals with AI-based models. However, their influence is limited to intersections. The road infrastructure on UAE is such that a majority of travel time is spent on 6-12 lane highways that connect different parts of the city and extends to other emirates, 100s of kilometers long. Instead of intersections, there are more flyovers and other structures to keep traffic flowing [13].

Reinforcement Learning has also been applied for optimal routing, to be used by drivers directly through apps, and have shown good results as well. There are already multiple solutions for this in the market, whereas VSL was comparatively less explored.

There are aspects of road design and making self-driving vehicles follow ideal behavior that contribute to this, but this research is focused on a solution authorities can take to assist traffic flow with existing infrastructure [13]. Road expansions are also not always feasible.

Some studies have experimented with lane-wise VSL, but these were not considered in this project due to its disadvantages – it has been observed that tendency to changing lanes & causing traffic breakdown increases when a driver is on a lane with lower limits than others [2].

1.4 Benefits of VSL

Studies demonstrate that VSL can effectively improve traffic flow and restore road capacity, i.e. the maximum number of vehicles the road can handle per hour, by limiting inflow into bottlenecks [5]. They've shown to increase induced greater road occupancy, driving speed at medium/high densities, decreased travel times from one point to another, some concluded with real-world data [2].

- Reduced Congestion – Many experiments have proven the effect of VSL in reducing congestion mainly through simulation results of various traffic models [1], [9].
- Safety - Bringing all vehicles to the same speed reduces the possibility of accidents, increasing safety of all drivers.
- Environmental Benefits – Studies have shown that consistent traffic flow & smoother speed pattern leads to emission reduction [2]. Addressing traffic congestion also contributes to SDG 13, Climate Action.

2. BACKGROUND

This section details past solutions to VSL, with their limitations.

These are some common parameters used to represent the extent of traffic at a point on the highway [1], [9]:

- Density - How close vehicles are. The critical density is the maximum the highway can handle.
- Flow – Number of vehicles passing through per unit time
- Speed – Average speed of vehicles
- Vehicle Hours – The time spent by a vehicle on the highway. The main requirement a driver cares about is how quickly they can get to their destination.
- Vehicle delay – Vehicle Hours minus the same during free flow/minimum time, i.e. delay

The characteristics of a highway can be represented by [9]:

- Capacity – How many vehicles the highway can take in. This is compared to demand - how many vehicles are coming in.
- Length – Length of the road stretch.

When there is congestion, there is reducing the distance between vehicles, and lower speeds are required to maintain safety.

To enhance coordination, VSL strategies are designed to optimize traffic throughput from a system-wide perspective over a specified time horizon. For any applied traffic control measures, such as VSL, it is crucial to ensure that their implementation does not significantly reduce traffic flow elsewhere in the network, which could negatively impact overall performance. To address this, VSL control can be designed with the objective of maximizing network efficiency or minimizing total time spent (TTS) within the system, including delays caused by queueing at on-ramps.

2.1 Rule-based Solutions

Early VSL systems were primarily designed using simple, reactive, rule-based logic. These systems adjusted speed limits based on preselected thresholds of traffic flow, occupancy, or speed [2].

Some of these were not focused on traffic flow, but specific aspects like emissions or safety – dealing with speed variance to reduce crash frequency, but at the cost of travel time [2]. This proposal will also explore a the addition of a safety aspect in the formulation.

The primary limitation of these early rule-based strategies lies in their reactive nature. Because they relied on real-time traffic data, there was often a lag in response, making them less effective in addressing rapidly changing traffic conditions. By the time VSL actions were implemented, traffic breakdowns might have already occurred, reducing their ability to resolve congestion. This highlights the need for more proactive, predictive VSL systems that can anticipate and mitigate traffic issues before they escalate [2].

2.2 Control-based Solutions

Advanced control strategies address the limitations of traditional rule-based approaches by predicting future traffic conditions and implementing proactive measures to prevent traffic breakdowns. These strategies aim to anticipate and mitigate congestion by reducing inflow into potentially jammed areas, thereby resolving shockwaves before they occur. Unlike reactive systems, advanced approaches eliminate the time lag between traffic changes and the optimization of control variables. The optimization process is repeated periodically using updated traffic data to improve demand predictions. Such closed-loop feedback control systems include components like monitoring, state estimation and prediction. These systems rely on continuous traffic data input, enabling real-time adjustments to control actions when discrepancies between predicted and actual traffic states are detected [2]. These approaches fall under 'Model Predictive Control' (MPC) strategies, which involves making decisions by using a model to predict future states. For example, it can predict the front distance a vehicle will reach if it accelerates for a specific amount of time, allowing the system to decide on the optimal acceleration to maintain an ideal front-back distance [].

Models like "SPECIALIST" effectively detect and resolve shockwaves, demonstrating high success rates in practical applications. Other advancements integrated dynamic demand and supply functions, enabling better representation of phenomena like capacity drops and shockwave formation. Some model the speed control variable as linear approximations. Some also incorporate sensors on vehicles to accurately track their movement, leading to a 'probe-based' system, but this is microscopic & unscalable [2].

2.3 Reinforcement Learning

Reinforcement learning (RL), on the other hand, is much more adaptive to dynamic environments compared to earlier solutions, and can handle high complexity. Reinforcement learning is a category of artificial intelligence & under machine learning, where the model or 'agent' is placed in an environment, and is given a set of possible actions. The target values or best actions are not given to the model, so the model learns through trial-and-error – where given a state, each action results in a 'reward', indicating whether that was a step in the right direction or not. As a result, the model comes up with state-action pairs, known as policies, that and the most optimal. There are 2 broad categories to achieve this – q-learning is one, that involves calculating the reward for each action given a state, and picking the one with the highest. The other is policy learning, where policies are directly learnt without explicitly calculating rewards.

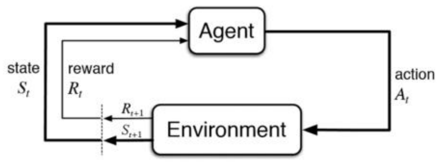

Fig. 1: Reinforcement Learning Flow

The first step to building a RL system is to formulate the problem as a Markov Decision Process (MDP), a method of modelling sequential decision making problems [9]. This would be defining the states and actions in the environment and connecting them – the start, all possible steps and routes it can go from there and how the state would change as a result of each, till finish. It can either be deterministic i.e. fixed outcomes, or probabilistic i.e. varying to some extent. There is an assumption involved here that the outcome only depends on the present state. MDPs that consider previous states are said to be ordered to the number of previous states considered plus one, and fall under a category called Partially Observable Markov Decision Process (POMDP). The states can be a set of finite values, or infinite continuous values. The reward or Q-function is calculated by adding 2 parts – the value of the current state, and the maximum rewarding state-action pair values, where we count the probability of getting that next state taking that action as well. The is also a 'discount factor' that represents how important the current reward over future ones is. Together, this is known as the Bellman equation [10].

An important study extensively reviewed for this project [9] uses reinforcement learning for VSL. For the MDP in [9], the highway is divided into sections where each need to be assigned a VSL. Each section will have a state that contains variables for its current density & speed, and the initial state is set to be the standard maximum limit. Action spaces were explicitly defined based on regulations. They were made dependent & restricted on the current state, to avoid drastic shifts in output speeds. However, this approach could over-constrain the system, potentially leading to suboptimal policies by ruling out useful actions in some scenarios.

[9] uses Q-learning, which comes under model-free RL. This means it does not 'model the full environment', and just populates the state-action table with rewards. It does not directly pick up on all the options it could take and where it would lead it (transition probabilities), which further steps will be presented after this step is taken, etc. It is simple, and preferred when the environment is fixed with fixed rewards, but involves more exploration. [9] justifies the use of Q-learning due to initially unknown rewards & probabilities. Function approximation was considered a viable choice due to large or continuous state spaces, and they decided on artificial neural networks for this task. However, below is a better solution that also deals with these issues, while offering significant advantages.

The reward function in [9] is defined based on the vehicle hours, with bigger punishments (negative rewards) for longer hours. There is a threshold defined such that no punishment is given when output speed is above it. This study also considers traffic predictions that many others don't, and demonstrates better yields with it. The state description was extended to include predicted density & speed. This project will also consider predictions.

2.4 Soft Actor-Critic

Deep reinforcement learning is the convergence of RL and deep learning, leveraging artificial neural networks (ANNs) for RL tasks. This combination enables the extraction of complex, high-dimensional nonlinear relationships from input data, making it particularly effective for decision-making in dynamic environments. The Soft Actor-Critic (SAC) algorithm, an advanced off-policy RL method (meaning it can learn from previously collected data/interactions rather than requiring fresh interactions with the environment for every update), is based on the maximum entropy framework (which encourages policies that are both high-performing and exploratory, reducing the risk of overly rigid or suboptimal strategies) [15]. SAC employs a stochastic actor-critic architecture (where the "actor" decides actions, and the "critic" evaluates them), featuring separate policy and value function networks. It benefits from entropy regularization (which keeps the policy from becoming too deterministic too quickly, encouraging better long-term exploration) to improve stability and sample efficiency (learning faster with fewer interactions).

In the context of Variable Speed Limits (VSL), reinforcement learning has been explored as a means of dynamically optimizing speed control strategies to mitigate congestion and enhance safety. Prior studies have applied SAC to VSL with a focus on mixed traffic environments that include Connected Autonomous Vehicles (CAVs) [14]. These approaches utilize image-based traffic observations, and deep reinforcement learning models to adjust speed limits in response to real-time traffic flow. However, our project diverges from this by applying SAC to standard VSL scenarios (without relying on CAVs).

This study implements SAC on the METANET traffic model, optimizing macroscopic VSL control parameters to assess their impact on overall traffic efficiency. Unlike prior work that emphasized microscopic traffic, our approach focuses on macroscopic state representations. This enables direct integration with traditional traffic flow models.

2.5 Modeling & Simulation

A significant amount of research has been conducted in the traffic flow domain, focusing on modeling and optimizing highway traffic flow. These can be classified into two categories: microscopic and macroscopic. Macroscopic prediction models remain the primary choice for VSL studies due to their ability to effectively capture aggregate traffic dynamics and simplify implementation across large-scale networks. While microscopic models show promise in representing individual driver behaviors with greater granularity, they are computationally demanding, and introduce complexities in modeling lane-changing behavior and multilane road networks.

METANET-based models have been widely used to simulate & test transportation networks, and the effect of applying various strategies to control it. With both synthetic & real-world data, these macroscopic models demonstrated their capability to accurately describe the dynamic behaviour of traffic and reproduce all relevant phenomena, highlighting METANET's effectiveness as a tool for traffic prediction and management [2]. They use averaged parameters that represent the traffic as a whole, which is acceptable for this use-case. We can set the duration time of the simulation and the step size at which models are applied to observe reactions.

There have been many modifications made to the original METANET to suit specific use-cases within traffic modelling, such as support for multiple lanes, vehicle types, etc. [9] also uses METANET, with a few changes. They consider flow, density & speed, with constants defining the length of the highway and no. of lanes, and check traffic volumes of on-ramps & off-ramps. Traffic near specific on-ramps compared to other lanes is a common sight everyday in the UAE. They divide the highway into sections, and observe the incoming & outgoing traffic volumes of both in each, and queue length of the ramp. They keep the maximum density (when speed nears 0) & ideal free flow speed as reference values. A step size is defined such that VSLs outputs are calculated only when the METANET step is a multiple of the size.

3. PROPOSAL

The aim of this project is to build an updated, scalable & efficient VSL system, trained & tested on real world data. This section has detailed the changes proposed compared to previous approaches, and the plan ahead to implement it.

3.1 Algorithm

Let’s start with getting the intuition. Looking at Fig. 1, we are looking at a highway, and need to assign speed limits based on current data. There is congestion on the second half, where there is an on-ramp with incoming vehicles. What should the VSLs be? A good approach would be to gradually reduce speeds closer to the average speed of the congested area. So first will be a little lesser than the standard limit, second will be even lesser, and so on. When should they be issued? Best if before the congestion happens, and avoid the buildup as much as possible. After this area, speeds can gradually rise, and this should happen as density shows signs of decreasing. This is how we want the model to work.

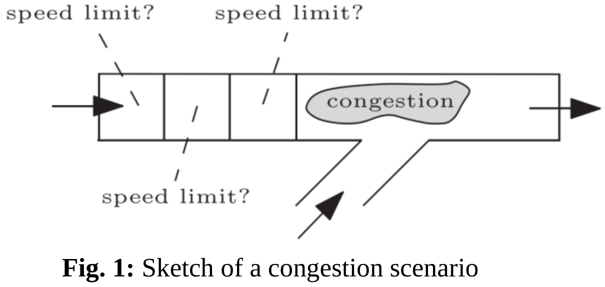

Fig. 1: Sketch of a congestion scenario

An MDP was formulated in [9], but predicted values were added with the current values together in the state descriptions. Predictions are not ground truth, so these should be considered partially observable, and the problem should be formulated as a POMDP.

The Soft Actor-Critic (SAC) algorithm is well-suited for optimizing Variable Speed Limits (VSL) due to its efficiency, stability, and adaptability in complex traffic environments. SAC is a highly sample-efficient reinforcement learning method, meaning it can learn effective speed control strategies with fewer interactions—an essential advantage given the high computational cost of traffic simulations. Unlike conventional RL methods that struggle with stability in high-dimensional environments like multi-lane highways with fluctuating traffic densities, SAC's twin Q-network architecture mitigates instability and ensures reliable learning. Moreover, SAC optimizes a stochastic policy, balancing exploration and exploitation to dynamically adjust VSL in response to changing traffic conditions. This is particularly valuable in uncertain environments, where driver behavior, congestion levels, and external factors (e.g., weather, road incidents) fluctuate. Although prior applications of SAC in VSL have focused on mixed traffic with Connected Autonomous Vehicles (CAVs), its fundamental strengths—handling stochasticity, optimizing continuous control, and learning stable policies—make it equally applicable to standard VSL systems. Even in purely human-driven traffic, SAC can harmonize speeds, reduce bottlenecks, and enhance overall throughput, making it a strong candidate for this study's proposed approach.

To enhance the safety of the reinforcement learning-based Variable Speed Limit (VSL) control, this study incorporates a safety-constrained SAC framework, inspired by Worst-Case Soft Actor-Critic **(WCSAC)** [34]. Traditional RL-based methods optimize policies based solely on reward maximization, potentially leading to unsafe actions. Instead, we introduce a safety critic, which operates parallel to the reward critic, estimating the risk of unsafe speed limit decisions. The safety metric is modeled using Conditional Value-at-Risk (CVaR)**,** which ensures that policies minimize worst-case traffic conditions, such as abrupt decelerations that could lead to collisions. The SAC algorithm is modified to include an adaptive safety weight, adjusting the trade-off between safety and efficiency dynamically. This ensures that speed recommendations harmonize traffic flow while maintaining a low probability of sudden braking events and excessive speed variations. By implementing this risk-aware SAC adaptation, the model can generate safe and stable speed limits under varying traffic conditions, making it more reliable for real-world deployment.

3.2 Simulation Framework

For this study, we adopt a macroscopic modelling approach due to its efficiency in capturing aggregate traffic dynamics and suitability for large-scale simulations. While microscopic models offer detailed representations of individual vehicle behaviours, they are computationally intensive and introduce challenges in modelling lane-changing and multi-lane interactions. Given the focus on optimizing VSL at a network-wide level, macroscopic models provide a balance between accuracy and computational feasibility. We utilize the METANET traffic model, a widely used macroscopic framework for simulating freeway traffic flow and evaluating control strategies. METANET represents traffic behaviour using averaged state variables such as speed, density, and flow, making it well-suited for VSL optimization. By defining appropriate simulation durations and step sizes, we can analyse the impact of dynamic speed limit adjustments on overall traffic performance.

To tailor METANET for our study, we implement modifications that enhance its applicability to the UAE traffic environment. Our model divides the highway into multiple segments, monitoring traffic flow, density, and speed at each. Unlike the prior adaptation that emphasize mixed traffic with autonomous vehicles, our approach focuses on conventional traffic with human drivers. We incorporate on-ramp and off-ramp interactions, as these significantly affect congestion in urban highways. Additionally, the step size for VSL updates is aligned with METANET's computational cycle to ensure smooth integration between the reinforcement learning controller and traffic simulation. The model references critical density thresholds, distinguishing between free-flow, congested, and near-gridlock conditions, allowing for adaptive speed limit interventions.

3.3 Data

To ensure that the METANET simulation accurately reflects real-world traffic conditions in Dubai, this study will collect real-time traffic data using the TomTom Traffic API. TomTom provides high-resolution traffic insights, including traffic flow (speed, congestion levels) and incident reports (accidents, roadblocks, construction delays)**.** The Traffic Flow service offers real-time observed speeds and travel times across Dubai's road network, while the Traffic Incidents service provides data on disruptions affecting vehicle movement. By leveraging TomTom's RESTful API endpoints, traffic data will be queried and integrated into the simulation, allowing for data-driven calibration of speed limits and reinforcement learning models. This approach ensures that the VSL optimization strategies are tested under conditions that closely mirror Dubai's actual traffic patterns.

3.4 Evaluation

To assess the effectiveness of the Soft Actor-Critic (SAC) algorithm in optimizing Variable Speed Limits (VSL), a multi-faceted evaluation approach will be employed. The performance of the learned policies will be analysed through traffic simulations on the METANET model, using real-world data from Dubai's road network, collected via the TomTom Traffic API.

1. Performance Metrics:
The evaluation will primarily focus on the following key metrics:

- **Total Time Spent (TTS):** Measures the mean time taken by vehicles to traverse the network.
- **Traffic Throughput:** Represents the number of vehicles passing a given point per unit time.
- **Speed Variability:** Analyses fluctuations in vehicle speeds to assess traffic stability.
- **Congestion Reduction:** Assesses the impact of VSL on bottleneck formations.
- **Safety Metric:** Evaluates potential accident risks based on speed harmonization and rapid deceleration rates.

2. Baseline Comparisons:
To determine the efficacy of the SAC-based VSL control, results will be compared against:

- **No Control (Fixed Speed Limits):** A scenario where speed limits remain constant.
- **Rule-Based VSL:** A traditional approach that adjusts speed limits based on predefined congestion thresholds.
- **Alternative RL Approaches:** Potential comparisons with other RL-based VSL methods from literature.

3. Simulation Scenarios:
Multiple traffic scenarios will be tested to ensure robustness, including:

- **Peak vs. Off-Peak Traffic:** Evaluating performance under varying congestion levels.
- **Incident-Induced Congestion:** Testing how SAC-VSL adapts to sudden disruptions.
- **On-Ramp & Off-Ramp Interactions:** Assessing how VSL influences merging and diverging traffic flows.

By leveraging data-driven simulations and comparative analysis, this evaluation framework aims to establish the practical feasibility of reinforcement learning-based VSL control in large-scale urban freeway networks.

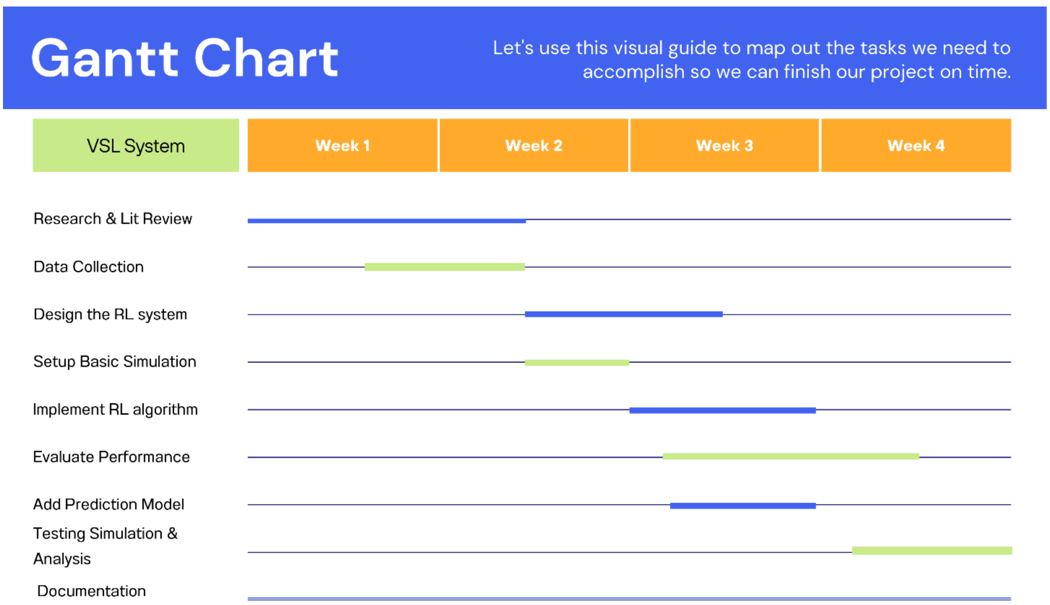

Fig. 2: Gantt Chart

4. FUTURE PROSPECTS

4.1 Potential Enhancements

Multi-agent soft actor-critic (MA-SAC) algorithm can be implemented for traffic signal control, where agents are deployed at each intersection to optimize traffic flow collectively.

The UAE utilizes a 'SCOOT' system, which is a centralized traffic management system equipped with sensors to monitor vehicle volumes. These sensors measure traffic flow, vehicle counts, and the average speeds. This data could be integrated into this VSL project to make decisions based on accurate real-time traffic data without the involvement of a third party [8].

4.2 Application Goals

This project will be presented to the road & Transport Authority (RTA), aiming to be implemented on the roads of UAE.

4.3 Other Considerations

A study of VSL results at different points of a road done in [3] is that it worsens congestion when placed right after an on-ramp. The solution to handling this would be to limit the number of vehicles leaving the ramp until road capacity is back to normal, a technique known as ‘ramp metering’.
Failing to consider the possibility of phantom jams occurring, leads to too many cars entering the circuit [6].

The effectiveness of VSL is highly dependent on drivers’ adherence to them [2]. UAE has been constantly advancing their government services, as well as their enforcement mechanisms. While cameras used to depend on manual checks & radar technologies, there are now mobile cameras on police cars, and cameras that observe speeds over time & average them. They’re using LIDAR sensors & inductive loops on roads, and fines are instantly applied and sent to violators via SMS. AI-powered analyses detect vehicle details & other safety aspects like presence of safety belts, phone usage, etc. These initiatives have significantly contributed to safer roads and better compliance [7]. Along with strict regulations and heavy fines, the UAE enforces a black points system, which can result in the confiscation of a driver's license, to promote compliance with traffic regulations [8]. The number of traffic deaths were reduced by 66% over the last decade through these proactive measures. The highways of Dubai also have Variable Message Signs alerting drivers of nearby traffic accidents and unstable weather conditions. With proven strong compliance, infrastructure and most importantly a thirst for innovative solutions, VSL is highly likely to positively impact this region.

CONCLUSION

The study demonstrates the potential of Variable Speed Limits (VSL) implemented using the Soft Actor-Critic (SAC) reinforcement learning algorithm to optimize traffic flow on long highway networks. By leveraging real traffic data from Dubai and simulating with the METANET traffic model, the proposed approach addresses key challenges in traffic congestion, safety, and environmental impact. The integration of a safety metric further enhances the system's practical applicability.

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